GEARBOx Output Discussion

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Current NER pipeline

- Instructor framework
- Convert enumerations to Enum class
- Force the GPT output items in the class only

```
class Diagnosis(Enum):
  APL = "Acute promyelocytic leukemia (APL)"
  AML = "Acute myeloid leukemia (AML)"
  DSAML = "Down syndrome acute myeloid leukemia (DS AML)"
  BALL = "B-cell acute lymphoblastic leukemia (B-ALL)"
  JMML = "Juvenile myelomonocytic leukemia (JMML)"
  MPAL = "Mixed-phenotype acute leukemia (MPAL)"
  MDS = "Myelodysplastic syndromes (MDS)"
  MDSAML = "Myelodysplastic syndrome (MDS) who have progressed to AML"
  TAML = "Treatment-related acute myeloid leukemia (tAML)"
  TALL = "T-cell acute lymphoblastic leukemia (T-ALL)"
  ALL = "Acute lymphoblastic leukemia (ALL)"
  KMT2AALL = "KMT2A acute lymphoblastic leukemia (KMT2A-ALL)"
  ETPALL = "Early T-cell precursor acute lymphoblastic leukemia (ETP-ALL)"
  ALAL = "Ambiguous lineage acute leukemia (ALAL)"

class Bm_recent_blast_method(Enum):
  MORPHOLOGY = "Morphology"
  FLOWCYTOMETRY = "Flow cytometry"

class Type_transplant(Enum):
  ALLOGENEIC = "Allogeneic"
  AUTOLOGOUS = "Autologous"

class Expression_alk_type(Enum):
  ALKR = "Rearrangement (ALK-R)"
  ALKA = "Amplification (ALK-A)"
  POINTMUTATION = "Point mutation"

class Yes_no(Enum):
  YES = "Yes"
  NO = "No"
```

Current NER pipeline

- Instructor framework
- Convert query to two parts:
  - Fact
  - Answer
- Incorporate examples with correct answer for complex sentences

```python
class Refractory_Relapse(BaseModel):
    ever_refractory_fact: str = Field(..., description='The fact that indicates whether the
    ever_refractory: Optional[Yes_no] = Field(
        None,
        description='Does the eligibility criteria mention refractory disease? Example: refrac

    count_refractory_fact: str = Field(..., description='The fact that indicates the number
    count_refractory: Optional[int] = Field(
        None,
        description='The number of occurrences of refractory disease are required for eligibil

    current_refractory_num_cycles_fact: str = Field(..., description='The fact that indicate
    current_refractory_num_cycles: Optional[int] = Field(
        None,
        description='What is the minimum number of cycles of therapy required for refractory c

    ever_relapse_fact: str = Field(..., description='The fact that indicates whether the eli
    ever_relapse: Optional[Yes_no] = Field(
        None,
        description='Is confirmed or suspected relapse disease mentioned in the eligibility cr

    count_relapse_fact: str = Field(..., description='The fact that indicates the minimum nu
    count_relapse: Optional[int] = Field(
        None,
        description='What is the minimum number of relapses required for eilgibility? Example

    model_config = ConfigDict(
        json_schema_extra={
            "examples": [
                {
                    "ever_refractory_fact": "Refractory leukemia, defined as persistent leuk
                    "ever_refractory": "Yes",
                    "count_refractory_fact": "Refractory leukemia, defined as persistent leu
                    "count_refractory": None,
                    "current_refractory_num_cycles_fact": "Refractory leukemia, defined as p
                    "current_refractory_num_cycles": 2,
                    "ever_relapse_fact": "Disease relapse or the presence of refractory dise
                    "ever_relapse": "Yes",
                    "count_relapse_fact": "Disease relapse or the presence of refractory dis
                    "count_relapse": 2
                },
            ]
        }
```

Consistency in Ontology File

- “None” and “Not applicable” should be simplified to “None”, which means can’t find related information from context.

Unclear Issues in Ontology File

- The meaning of “No” and “None” is unclear.
  - Does the eligibility criteria **mention** a diagnosis of isolated extramedullary disease (i.e. the absence of bone marrow involvement)?
    - If a “diagnosis of isolated extramedullary disease” is mentioned, we expected output of “Yes”,
    - If a “diagnosis of isolated extramedullary disease” isn’t mentioned, what is the expected output from model?
    - The answer of “No” is reasonable since the context don’t mention it. However, the answer of “None” is also reasonable since we can’t find related information from the context.

Solutions?- #Example1

- Goal: avoid mixture output of No and None; handle wash-out variables

- Does the eligibility criteria mention a diagnosis of isolated extramedullary disease (i.e. the absence of bone marrow involvement)?

  - **Pipeline development:**
  - 1. Extract the medical terms - **Keyword search**
  - 2. If exist, extract context info - **Semantic Search**

  - **Ontology file refining – Need Jooho's help**
  - 4. Provide medical terms: isolated extramedullary disease, bone marrow involvement
  - 3. Design query based on GEARBOx application: *Does the criteria allows a diagnosis of isolated extramedullary disease (i.e. the absence of bone marrow involvement)?*

Solutions?- #Example 2

- Does the eligibility criteria mention any prior exposure to myelosuppressive anti-cancer agents?

  - 1. Extract the medical terms - **Keyword search**
  - 2. If exist, extract context info - **Semantic Search**
  - 4. Provide medical terms: myelosuppressive anti-cancer agents
  - 3. Design query based on GEARBOx application: *What situations or conditions allow prior exposure to myelosuppressive anti-cancer agents for inclusion in the eligibility criteria?*